# README

This folder contains the MATLAB programs used for learning, verification, and figure generation in the manuscript.

## Folder Structure

The code is organized into three main folders:

```
Learning/
Verification/
Figure_code/
```

### 1. Learning

The folder Learning contains the training program.

To run the learning process, open the folder Learning in MATLAB and execute:

```
main.m
```

Running this file will train the proposed controller and generate the final learned switching policy. The learned switching policy is then used in the verification stage.

### 2. Verification

The folder Verification contains the verification program.

After completing the learning process, open the folder Verification in MATLAB and execute:

```
main.m
```

This program uses the learned switching policy obtained from the Learning stage and produces the corresponding verification results.

The comparison results reported in the manuscript can be reproduced by modifying the training conditions according to the settings described in the paper and then running the learning and verification programs again.

### 3. Figure_code

The folder Figure_code contains the scripts used to generate the figures in the manuscript.

The figure-generation programs are organized as follows:

```
Figure_code/
  Fig.3/
    main.m
  Fig.4/
    main.m
```

```
Fig.5/
  main.m
Fig.6/
  main.m
```

To reproduce each figure, open the corresponding figure folder in MATLAB and run:

```
main.m
```

For example, to reproduce Fig. 3, open the folder:

```
Figure_code/Fig.3/
```

and execute:

```
main.m
```

The same procedure applies to `Fig.4`, `Fig.5`, and `Fig.6`.

## Recommended Running Procedure

The recommended procedure is:

1. Run `Learning/main.m` to obtain the learned switching policy.
2. Run `Verification/main.m` to verify the learned policy.
3. Run the corresponding `main.m` files in `Figure_code/Fig.3`, `Figure_code/Fig.4`, `Figure_code/Fig.5`, and `Figure_code/Fig.6` to generate the figures.